

Dancing with the Stars: Innovation Through Interactions

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Core Research Question

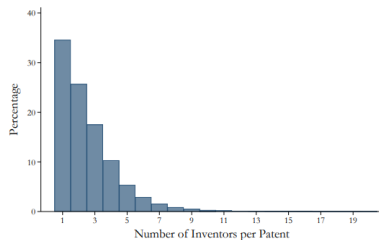
Main Goal

Micro-found innovation in research teams and embed **knowledge diffusion** (human capital formation) into an **innovation-based endogenous growth model**.

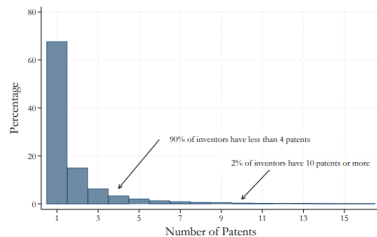
- **Key Insight:** Knowledge diffuses through **direct team interactions** and **external learning** (learning-by-doing).
- **Key Theoretical Contributions:**
 - Micro-found the idea production function at the team level.
 - Endogenize occupational choices (Team Leaders vs. Team Members).
 - Nest Kolmogorov Forward Equations inside a Balanced Growth Path.

Empirical Facts (EPO Patent Data)

Team Size Distribution



Patents per Inventor



Model Theory 1: Final Goods & Market for Ideas

Final Good Producers

Competitive producers $i \in [0, 1]$ using unskilled labor L_i and TFP $A_i(t)$:

$$y_i(t) = A_i(t)^\alpha L_i(t)^{1-\alpha} \implies Y(t) = \bar{A}(t)^\alpha$$

- **Buying Innovations:** Producers buy ideas of quality q to increase $A_i(t) \rightarrow A_i(t) + q$.
- **Value Function of Producers:** Linear in productivity $A_i(t)$:

$$V_i(A_i) = v \frac{A_i(t)}{\bar{A}(t)^{1-\alpha}} + \bar{v} \bar{A}(t)^\alpha, \quad \text{where } v = \frac{\alpha}{r + \frac{1-\alpha}{\alpha} g_w}$$

- **Idea Price (p):** Constant marginal value $\implies$ Idea price per unit of quality q :

$$p(t) = \beta \frac{v}{\bar{A}(t)^{1-\alpha}}$$

Model Theory 2: Occupational Roles (Leaders vs. Members)

Endogenous Role Division

Ideas for new patents arrive to a fraction ι of researchers each period.

■ Team Leaders:

- Have a new idea AND productivity $z > \hat{z}(s, t)$.
- Hire members, direct research, and collect profits $\pi(z, t)$.
- *Learning*: Learn **only externally** (m_X) outside the team.

■ Team Members:

- Have no new idea OR productivity $z \leq \hat{z}(s, t)$.
- Provide unit labor for wage $w(t)$.
- *Learning*: Learn **within teams** from Leaders (m) + external (m_X).

Model Theory 3: Team Production & Profits

- **Team Production Function:** A leader with knowledge z hiring n members produces an idea of quality q :

$$q = z^{1-\eta} n^\eta \quad (\eta \in [0, 1] \text{ is span of control})$$

- **Leader's Static Profit Maximization:**

$$\max_{n \geq 0} \{p(t) z^{1-\eta} n^\eta - w(t) n\}$$

- **Optimal Team Size (n), Quality (q), and Profits (π):**

$$n(z, t) = \left(\frac{p(t)\eta}{w(t)} \right)^{\frac{1}{1-\eta}} z, \quad \pi(z, t) = p(t) \left(\frac{p(t)\eta}{w(t)} \right)^{\frac{\eta}{1-\eta}} (1-\eta)z$$

More knowledgeable leaders run larger teams and earn higher profits.

HJB Equations

Team Leader's Value $V_l(z, s, t)$

$$\rho V_l = \pi(z, t) + m_X \int_z^\infty \Delta V_l(\tilde{z}) e(\tilde{z}, s, t) d\tilde{z} + (1 - \iota)(V_m - V_l) - \delta V_l + \frac{\partial V_l}{\partial t}$$

Team Member's Value $V_m(z, s, t)$

$$\rho V_m = w(t) + \max_m \left\{ m \int_z^\infty \Delta V_m(\tilde{z}) f(\tilde{z}, t) d\tilde{z} - C(m, s, t) \right\} + m_X \int_z^\infty \dots - \delta V_m + \frac{\partial V_m}{\partial t}$$

- **Interaction Choice:** Members choose meeting rate m at quadratic cost $C(m, s, t) = \frac{\kappa(s)}{2} m^2$ to acquire knowledge from Stars.

Model Theory 5: Cutoffs & Labor Market Clearing

- **Occupational Choice Cutoff $\hat{z}(s, t)$:** Indifference condition determines age-dependent threshold:

$$V_m(\hat{z}(s, t), s, t) = V_l(\hat{z}(s, t), s, t)$$

- **Labor Market Clearing (Determining $w(t)$):** Supply of members (below cutoff + fraction $1 - \iota$ above) equals Demand from leaders:

$$\begin{aligned} & \int_0^\infty [\Phi(\hat{z}(s, t)) + (1 - \iota)(1 - \Phi(\hat{z}(s, t)))] d\Omega(s) \\ &= \iota \int_0^\infty \int_{\hat{z}(s)}^\infty n(z, t) f(z, t) dz d\Omega(s) \end{aligned}$$

Model Theory 6: Knowledge Accumulation Dynamics

KFE for Team Members ($z < \hat{z}(s, t)$)

Evolution of productivity density $h(z, s, t_b)$ over age s :

$$\frac{\partial h}{\partial s} = \underbrace{-m(z, s)(1 - F(z, t))h + m(z, s)H(z, s, t_b)f(z, t)}_{\text{Inflows/Outflows from Team Interactions}} + \underbrace{-m_X(1 - E(z, s, t))h + m_X H(z, s, t_b)e(z, s, t)}_{\text{External Learning}}$$

- **Meetings Mechanism:** When member i meets j , knowledge updates: $z_i(t + \Delta t) = \max\{z_i, z_j\}$.
- **Cross-Sectional Distribution:** Aggregated over age distribution $\Omega(s) = 1 - e^{-\delta s}$: $F(z, t) = \int_0^\infty H(z, s, t)d\Omega(s)$.

Model Theory 7: Balanced Growth Path (BGP)

- **Detrended Variables:** $x = ze^{-gt}$. Quantiles of distributions grow at constant rate g .
- **Optimal Meeting Rate $m^*(x, s)$ along BGP:**

$$m^*(x, s) = \frac{1}{\kappa(s)} \int_{\max\{x, \hat{x}(s)\}}^{\infty} [\iota v_l(\tilde{x}, s) + (1 - \iota) v_m(\tilde{x}, s) - v_m(x, s)] \phi(\tilde{x}) d\tilde{x}$$

- **Aggregate Productivity Growth Rate (g_A):**

$$g_A = (1 - \eta)g, \quad \text{where } g = \int k_i dF(k_i)$$

- **Special Case (No Team Structure, Fixed m):**

$$g = (m + m_X)\theta \quad (\theta = \text{Pareto tail thickness})$$

Quantitative Strategy: SMM & Estimation

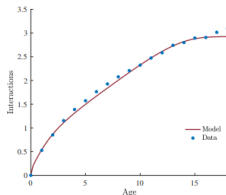
- Estimated via **Simulated Method of Moments** matching **87 data moments** using 12 structural parameters:

$$\chi := (m_X, \lambda, \theta, \nu, \eta, \delta, k_0/k, \alpha, \iota, \kappa_0, \kappa_1, \kappa_{\min})$$

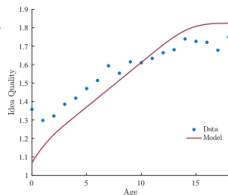
Parameter	Description	Value
m_X	External learning arrival rate	0.524
θ	Tail parameter of productivity distributions	0.271
η	Team member output elasticity ($1 - \eta = 0.95$ leader control)	0.048
δ	Death / retirement hazard rate	0.319
κ_0	Interaction cost function intercept	0.601
α	Final good elasticity w.r.t. aggregate TFP	0.121

- **Implied Rates:** Average endogenous meeting rate $m^* = 1.18 > m_X = 0.52$.

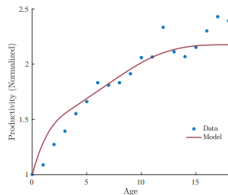
Model Fit: Targeted Moments (Figure 5)



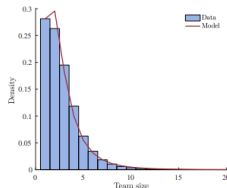
(A) HIGH-QUALITY INTERACTIONS



(B) IDEA QUALITY q



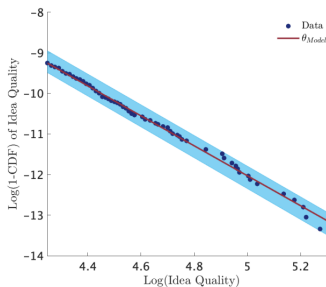
(C) PRODUCTIVITY GROWTH OVER INVENTORS' LIFE CYCLES



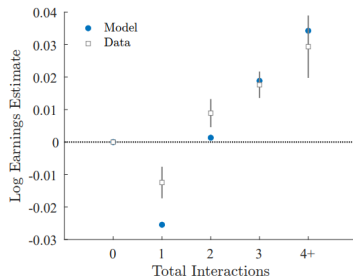
(D) TEAM SIZE DISTRIBUTION

Model Validation: Untargeted Moments

Tail of Idea Quality (> 99.9 th %ile)



Inventor Earnings & Interactions



Conclusion

- **Micro-Founding Growth:** Successfully links endogenous growth theory with rich micro panel data on patenting teams.
- **Stars Drive Human Capital:** Collaborating with high-knowledge co-inventors permanently pushes inventor productivity upward.
- **Core Takeaway:** Knowledge accumulation is a structured, team-based matching process. Team interactions are the main engine driving aggregate TFP growth.

Thank you!